UNIT 1 · 8 hours

Transfer Function and Mechanical Translational Systems



Introduction and Mathematical Modelling of Physical Systems

Control Systems (22EEG53)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

- Define the transfer function and state its properties
- Write the differential equations of mass-spring-damper systems
- Derive transfer functions of translational systems
- Find natural frequency and damping ratio from the model

Syllabus context

Introduction and mathematical modelling of physical systems: open loop and closed loop systems, examples, control system components. Transfer function of physical systems: mechanical systems (translational and rotational), electrical systems, analogous systems, block diagram reduction techniques, signal flow graphs.

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Transfer function

Definition: G(s) = L{output} / L{input} with all initial conditions zero.

Applies to linear time-invariant (LTI) systems.

Independent of input; depends only on system parameters.

Poles and zeros: roots of denominator / numerator; poles decide stability and response.

Characteristic equation: denominator of the closed-loop transfer function = 0.

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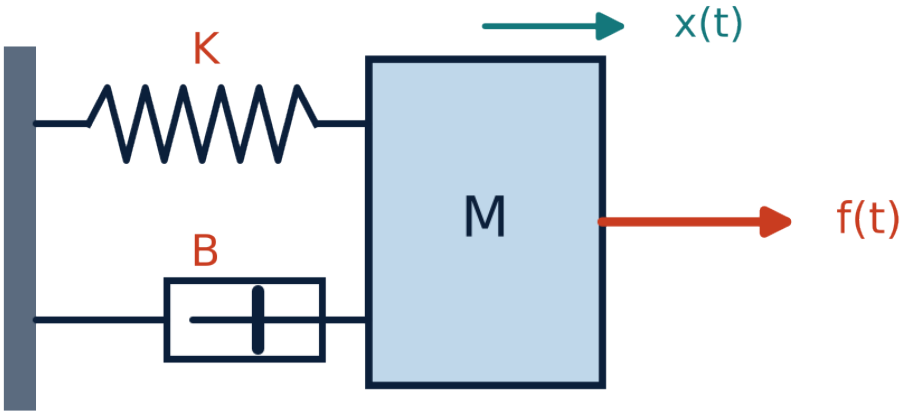
Translational elements

Element	Force relation	Laplace
Mass M	$f = M\, d^2x/dt^2$	$M\, s^2\, X(s)$
Viscous damper B	$f = B\, dx/dt$	$B\, s\, X(s)$
Spring K	$f = K\, x$	$K\, X(s)$

D'Alembert: the algebraic sum of forces on a mass (including inertia force) is zero.

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Mass–spring–damper system



- Free body diagram of M: $f(t) - B\,\dot{x} - K\,x = M\,\ddot{x}$.
- $M\,\ddot{x} + B\,\dot{x} + K\,x = f(t)$.
- Laplace (zero initial conditions):
- $(M\,s^2 + B\,s + K)\,X(s) = F(s)$.

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Transfer function and parameters

$$X(s)/F(s) = 1/(M s^2 + B s + K)$$

displacement/force

$$\omega n = \sqrt{(K/M)}$$

natural frequency

$$\zeta = B / (2\sqrt{(KM)})$$

damping ratio

- Standard form: $X/F = (1/K)\ \omega n^2 / (s^2 + 2\zeta\omega n\ s + \omega n^2)$.
- Static deflection = 1/K for a unit step force.

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Numerical: single mass

Problem M = 1 kg, B = 3 N·s/m, K = 2 N/m. Find X(s)/F(s), the poles, ωn and ζ.

- $X/F = 1/(s^2 + 3s + 2) = 1/[(s + 1)(s + 2)]$
- Poles at $s = -1, -2$ (real, distinct)
- $\omega n = \sqrt{2} = 1.414$ rad/s; $\zeta = 3/(2\sqrt{2}) = 1.06$

Answer: Overdamped (ζ > 1); poles −1 and −2

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Procedure for multi-mass systems

1

Mark displacements

Assign x1, x2, … to each mass.

2

Draw free-body diagram

For each mass draw all forces: springs, dampers, external.

3

Write equations

$\Sigma F = M\ddot{x}$ for each mass.

4

Laplace and solve

Solve simultaneous algebraic equations for the required ratio.

Two masses: $(M1s^2 + B1s + K1 + K12)\,X1 - K12\,X2 = F; \quad -K12\,X1 + (M2s^2 + B2s + K12)\,X2 = 0$

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Numerical: two-mass transfer function

Problem Masses M1, M2 joined by spring K12; force F acts on M1; no walls or dampers. Find X2/F.

- $(M1\ s^2 + K12)\ X1 - K12\ X2 = F$
- $-K12\ X1 + (M2\ s^2 + K12)\ X2 = 0 \Rightarrow X1 = (M2\ s^2 + K12)\ X2/K12$
- Substitute: $[(M1s^2 + K12)(M2s^2 + K12) - K12^2]\ X2 / K12 = F$
- $X2/F = K12 / [s^2(M1M2\ s^2 + K12(M1 + M2))]$

Answer: $X2/F = K12 / \{ s^2 [M1M2\ s^2 + K12\ (M1 + M2)] \}$

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Key takeaways

- Transfer function = L(output)/L(input) with zero initial conditions
- Mass, damper, spring: f = M$\ddot{x}$, B$\dot{x}$, Kx
- M s^2 + B s + K gives $\omega n = \sqrt{(K/M)}$, $\zeta = B/(2\sqrt{(KM)})$
- Write one force equation per mass

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Practice questions

Q1. Derive X(s)/F(s) for a mass–spring–damper.

Q2. M = 2 kg, B = 4, K = 8: find ωn and ζ.

Q3. Write equations for two masses connected in series with springs.

Q4. What are the properties of a transfer function?